

Main Menu

Welcome to the Smart City Critical Event System!

Smart City Critical Event System

1. Events
2. Updates
3. Queries
4. Metrics
0. Exit

Option: █

Main Menu → Events

Events Menu

1. Register new event
2. Remove resolved event
3. Search event by ID
0. Back

Main Menu → Events → Register new event

Event ID: 1
Severity (1-5): 1
Event type:
 1. Traffic Accident
 2. Traffic Light Failure
 3. Power Outage
 4. Flooding
 5. Fire
Type: 1
Region: North
Day: 3
Month: 04
Year: 2026
Hour: 16
Minute: 09
[OK] Event 1 registered successfully.

Main Menu → Updates

Updates Menu

1. Mark event as resolved
2. Update event severity
0. Back

Main Menu → Updates → Update event severity

Event ID: 1
New severity (1-5): 5
[OK] Event 1 severity updated to 5.

Main Menu → Queries

Queries Menu
1. Events by severity range 2. Events by region 3. Events by ID range 0. Back

Main Menu → Queries → Events by region

```
--- Active events in region: North ---
+-----+
| ID      : 1  
| Type    : Traffic Accident  
| Severity : 5  
| Region  : North  
| Status  : Active  
| Date    : 03/04/2026  
| Time    : 16:09  
+-----+
1 event(s) found.
```

Main Menu → Metrics

Metrics Menu
1. Show all metrics 0. Back

Main Menu → Metrics → Show all metrics

```
--- AVL Tree Metrics ---
Total height      : 1
Total nodes       : 1
Active events     : 1
Avg. balance factor : 0.00
Total rotations   : 0
```

Main Menu → Metrics → Show all metrics (tree with ID 2 added)

```
--- AVL Tree Metrics ---
Total height      : 2
Total nodes       : 2
Active events     : 2
Avg. balance factor : 0.50
Total rotations   : 0
```

Main Menu → Metrics → Show all metrics (tree with ID 3 added)

```
--- AVL Tree Metrics ---
Total height      : 2
Total nodes       : 3
Active events     : 3
Avg. balance factor : 0.00
Total rotations   : 1
```